Robot Vision and Learning
— From a Perspective of Embodied AI

Lecture 2 - Robotics I

Prof. He Wang

Course Organization: Intro to Robotics

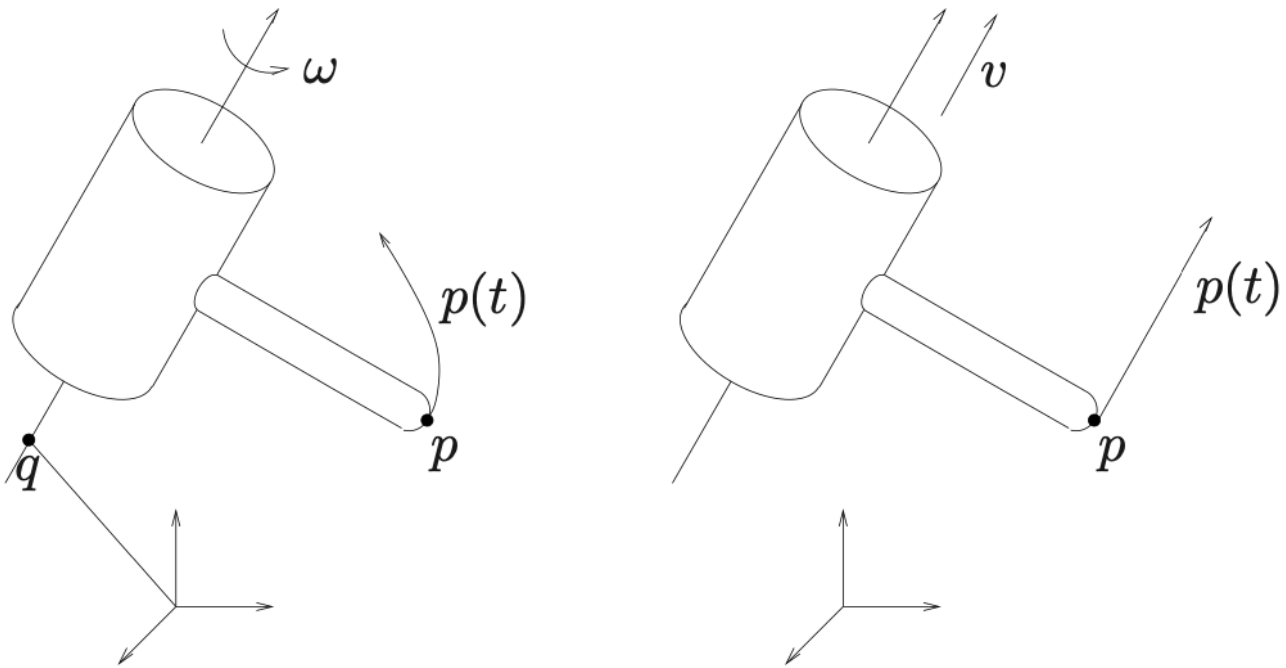
Lecture	**09/18/2024 15:10 Wednesday**	Basics of Robotics I	Kinematics, Rigid Transformation, Multi-Link Rigid-Body Geometry
Lecture	**09/25/2024 15:10 Wednesday**	Basics of Robotics II	Kinematics, Quaternion, SO(3)
Lecture	**10/09/2024 15:10 Wednesday**	Basics of Robotics III	Motion Planning, Control

(Tentative schedule)

For the complete and up-to-date schedule, see https://hughw19.github.io/RVAL/schedule/.

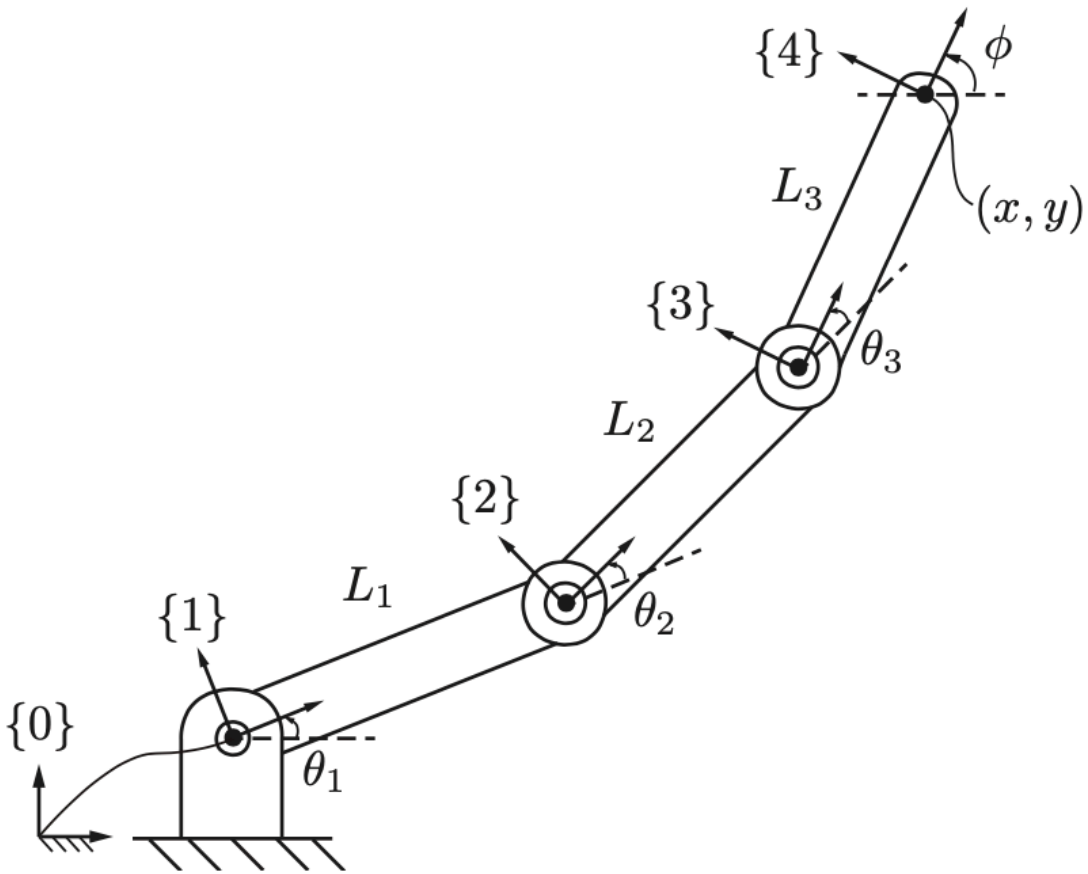
Topics Covered

- Modeling Robots by Rigid-Body Geometry



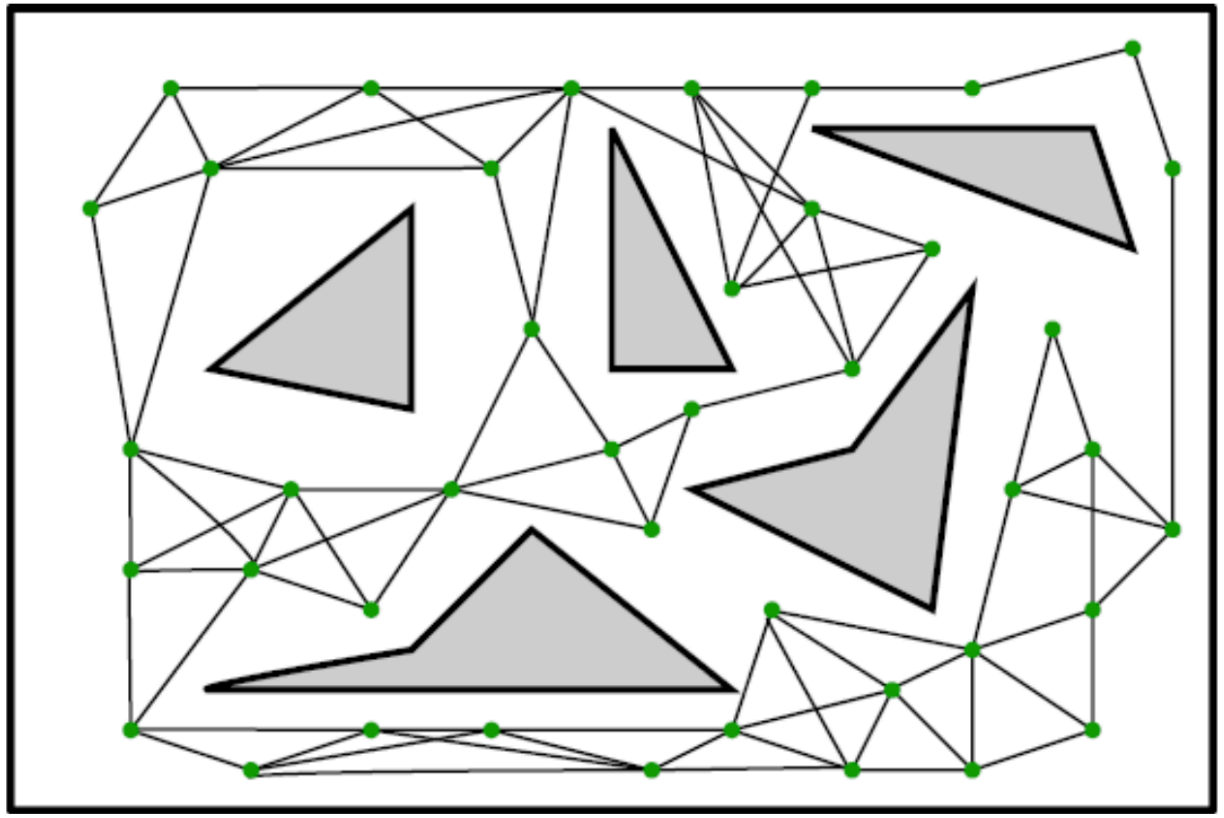
Topics Covered

- Forward and Inverse Kinematics of Robots



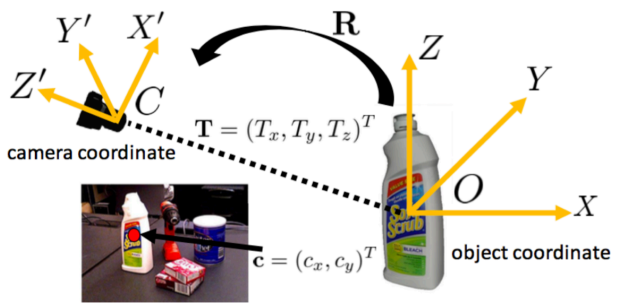
Topics Covered

- Classical Planning and Control



The Basics of Robotics

- **Kinematics (today's topic)**
- Dynamics
- Motion Planning
- Control



Kinematics vs. Dynamics

- Kinematics: describing the motion of bodies (position, linear/angular velocity, linear/angular acceleration, etc.)
- Kinematics **does not consider** how to achieve motion via force

- On the contrary, dynamics models all the way from force and torque to the motion

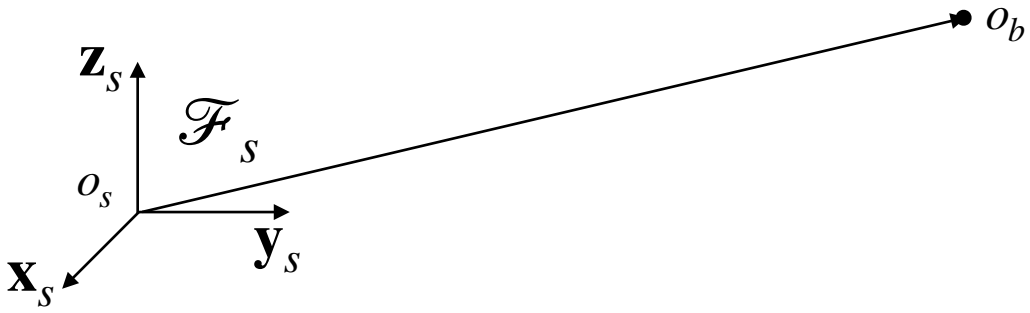
Rigid Transformation

Slides mainly borrowed from UCSD CS291-E00

Agenda

- Rigid Transformation
- $(R_{s\to b}, \mathbf{t}_{s\to b})$ for Coordinate Transformation
- $(R_{s\to b}, \mathbf{t}_{s\to b})$ as a Linear Transformation

Notation Convention



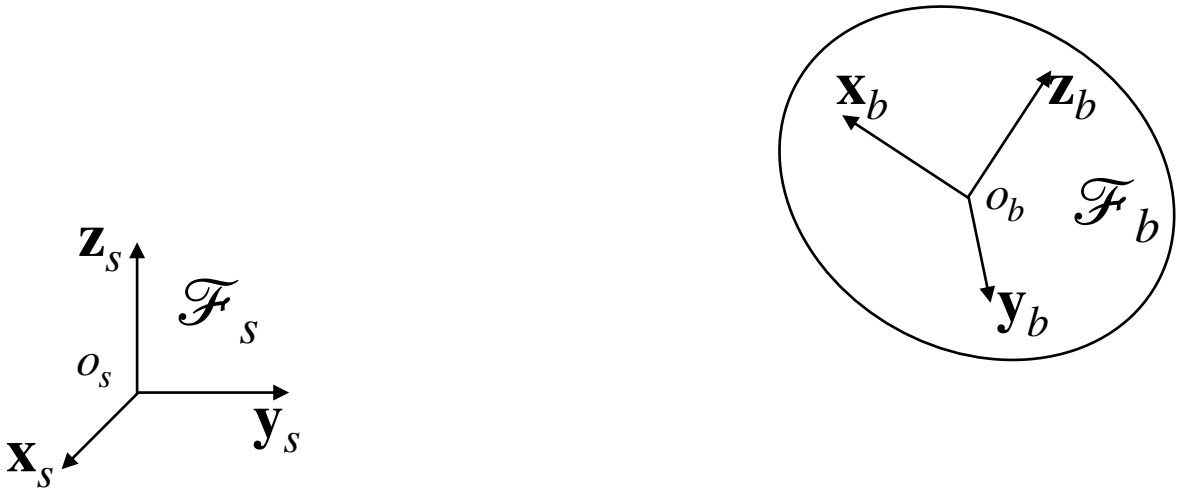
- An observer **records** the position of any point in the space **using a frame** $\mathcal{F}_s$
- We use ordinary letters to denote points (e.g., p), and bold letters to dente **vectors (e.g., $\mathbf{v}$)**
- When **writing equations**, we add a superscript to symbols to denote the recording frame, e.g.,

$$o_b^s = o_s^s + \mathbf{t}_{s \to b}^s$$

Rigid Transformations

- There is a rigid object, to which we bind a frame $\mathcal{F}_b$ (body frame) tightly, so that $\mathcal{F}_b$ moves along with the object

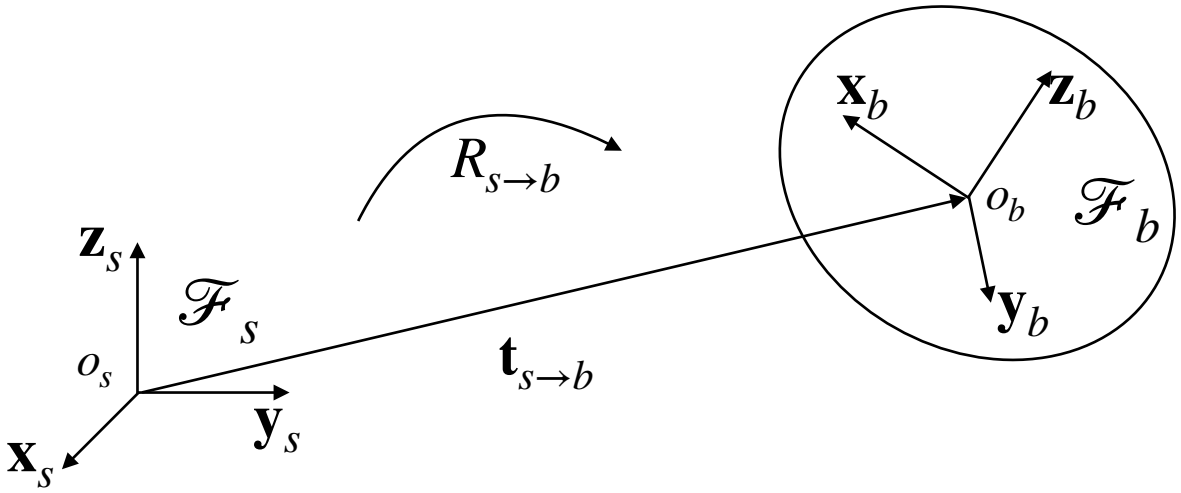
Rigid Transformations



- When talking about the pose of the *rigid* object, we ask:

  How to **transform** $\mathcal{F}_s$ so that it overlaps with $\mathcal{F}_b$?

Rigid Transformations

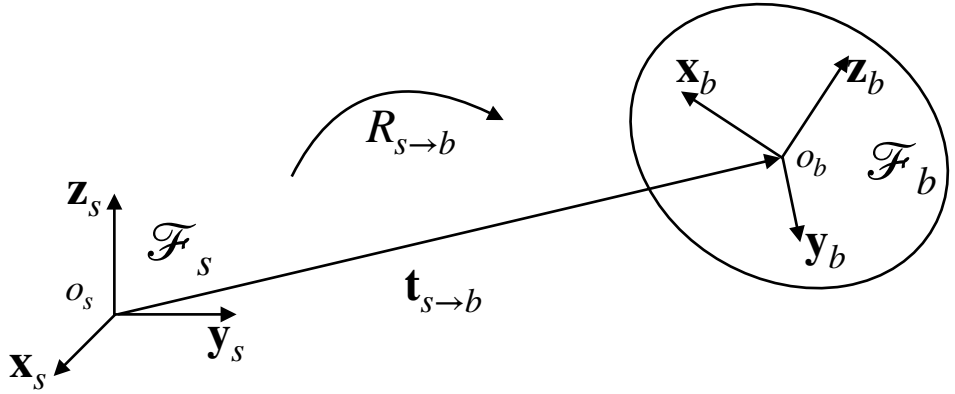


- We first translate $\mathcal{F}_s$ by $\mathbf{t}_{s\to b}$ to align o_s and o_b
- And then rotate by $R_{s\to b}$ to align $\{\mathbf{x}_i, \mathbf{y}_i, \mathbf{z}_i\}$ $(i = s \text{ or } b)$

Rigid Transformations

- Formally,
  - $o_b^s = o_s^s + \mathbf{t}_{s\to b}^s$
  - $[\mathbf{x}_b^s, \mathbf{y}_b^s, \mathbf{z}_b^s] = R_{s\to b}^s [\mathbf{x}_s^s, \mathbf{y}_s^s, \mathbf{z}_s^s]$

- Since the observer records everything using $\mathscr{F}_s$,
  - $o_s^s = 0$
  - $[\mathbf{x}_s^s, \mathbf{y}_s^s, \mathbf{z}_s^s] = I_{3\times 3}$

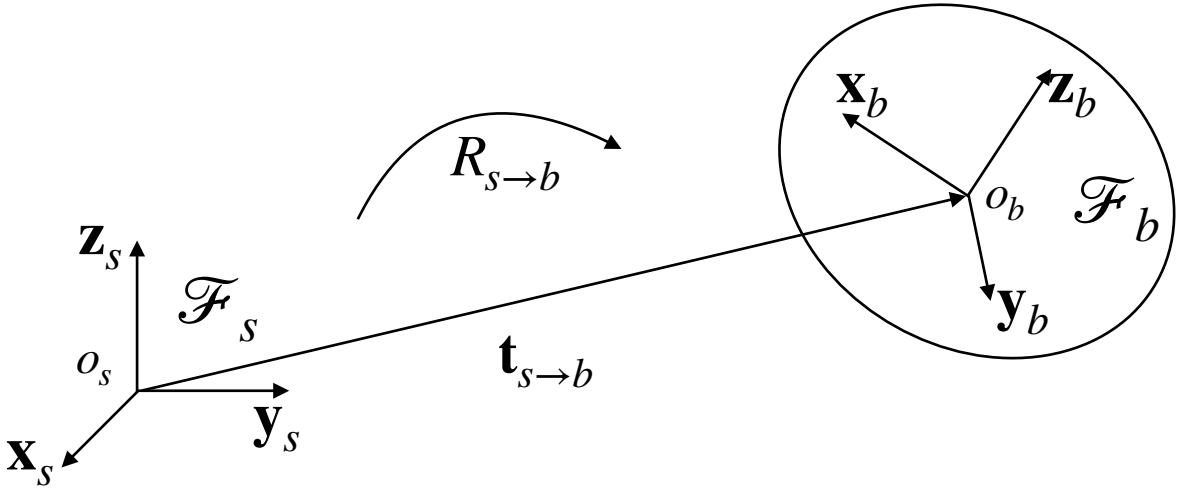


- Therefore,
  - $\mathbf{t}_{s\to b}^s = o_b^s$
  - $R_{s\to b}^s = [\mathbf{x}_b^s, \mathbf{y}_b^s, \mathbf{z}_b^s] \in \mathbb{R}^{3\times 3}$

Rigid Transformations

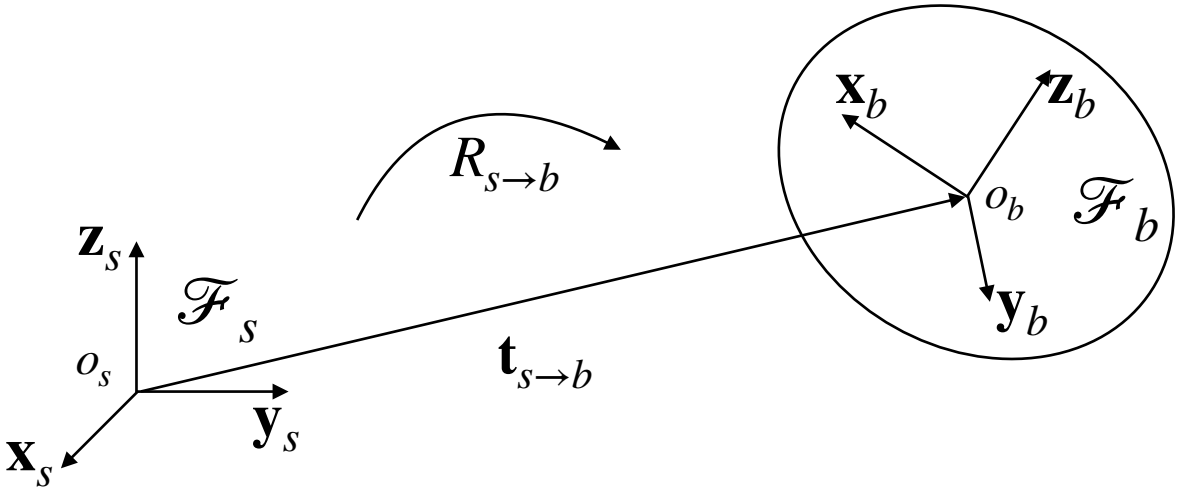
$(R_{s \to b}, \mathbf{t}_{s \to b})$ **for Coordinate Transformation**

Use Coordinate Transformation to Relate Coordinates in Frames



- Assume a second observer that records coordinates by $\mathcal{F}_b$
- Assume a point p on the body. Since $\mathcal{F}_b$ moves along the body, its coordinate recorded in $\mathcal{F}_b$, denoted as p^b, should **never change**.

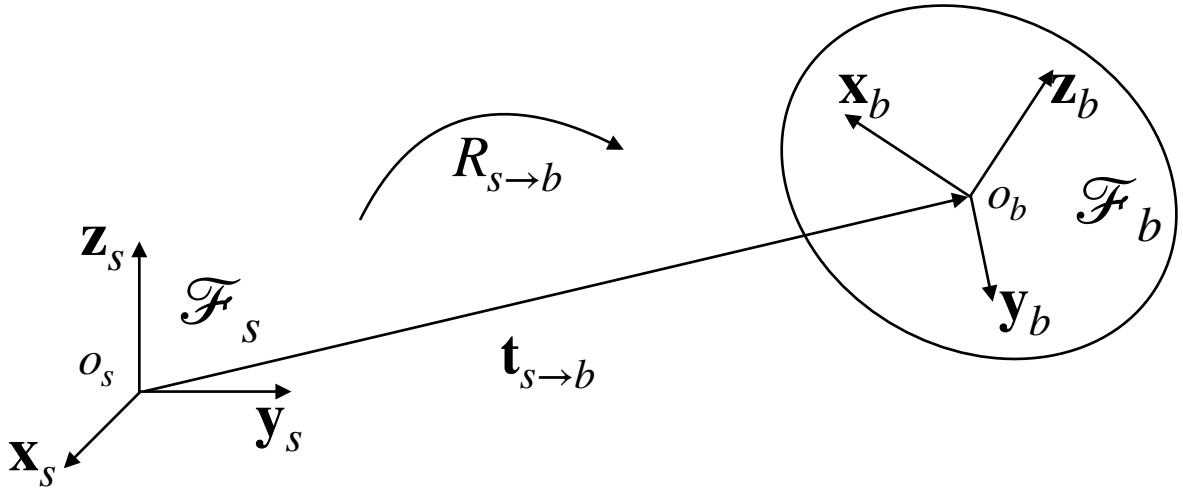
$(R_{s\to b}, \mathbf{t}_{s\to b})$ for Coordinate Transformation



- Imagine a process: $\mathcal{F}_b$ moves from $\mathcal{F}_s$ to the current location. This is how we define $(R^s_{s\to b}, \mathbf{t}^s_{s\to b})$.
- Since p moves along $\mathcal{F}_b$, it is moved from the **initial position**, $p^s = p^b$, to the current location:

$$p^s = R^s_{s\to b} p^b + \mathbf{t}^s_{s\to b}$$

$(R_{s\to b}, \mathbf{t}_{s\to b})$ for Coordinate Transformation



- $(R_{s\to b}, \mathbf{t}_{s\to b})$ transforms any **point** in the *whole space* by the following equation:

$$x'^s = R^s_{s\to b} x^s + \mathbf{t}^s_{s\to b}$$

Linear Transformation

- A linear transformation between two vector spaces V and W is a map $T : V \to W$ such that the following hold:
- $T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2)$ for $\forall \mathbf{v}_1, \mathbf{v}_2 \in V$
- $T(\alpha \mathbf{v}) = \alpha T(\mathbf{v})$

- We know that we can use matrix to describe such linear transformation and use matrix multiplication to perform these linear operations.

$(R_{s\to b}, \mathbf{t}_{s\to b})$ is not Linear

- $p_1^s = R_{s\to b}^s p_1^b + \mathbf{t}_{s\to b}^s$
- $p_2^s = R_{s\to b}^s p_2^b + \mathbf{t}_{s\to b}^s$
- $p_1^s + p_2^s \neq R_{s\to b}^s (p_1^b + p_2^b) + \mathbf{t}_{s\to b}^s$ when $\mathbf{t}_{s\to b}^s \neq \mathbf{0}$

- Similarly, $\alpha p_1^s \neq R_{s\to b}^s (\alpha p_1^b) + \mathbf{t}_{s\to b}^s$ when $\mathbf{t}_{s\to b}^s \neq \mathbf{0}$

- Can we represent this transformation as a linear operation?

Homogenous Coordinates

- Homogeneous coordinate for 3D Space:

$$\tilde{x} := \begin{bmatrix} x \\ 1 \end{bmatrix} \in \mathbb{R}^4$$

- Homogeneous transformation matrix:

$$T^s_{s\to b} = \begin{bmatrix} R^s_{s\to b} & \mathbf{t}^s_{s\to b} \\ 0 & 1 \end{bmatrix}$$

- Coordinate transformation under linear form:

$$\tilde{x}^s = T^s_{s\to b}\tilde{x}^b$$

- Ignore ˜ for simplicity in the future.

Homogenous Coordinates

- The coordinate transformation works for any choice of $\mathscr{F}_s$ and $\mathscr{F}_b$
- As a general rule, we have:

$$x^1 = T^1_{1\to 2} x^2$$

Some Rules of Homogenous Coordinate Transformation

By $x^1 = T^1_{1\to 2} x^2$, we have $x^2 = T^2_{2\to 1} x^1$ and $x^3 = T^3_{3\to 2} x^2$.

Therefore, $x^3 = T^3_{3\to 2} T^2_{2\to 1} x^1$. But $x^3 = T^3_{3\to 1} x^1$

- Composition rule: $T^3_{3\to 1} = T^3_{3\to 2} T^2_{2\to 1}$

By $x^1 = T^1_{1\to 2} x^2$, we have $x^2 = (T^1_{1\to 2})^{-1} x^1$

- Change of observer's frame: $T^2_{2\to 1} = \left(T^1_{1\to 2}\right)^{-1}$

Multi-Link Rigid-Body Geometry

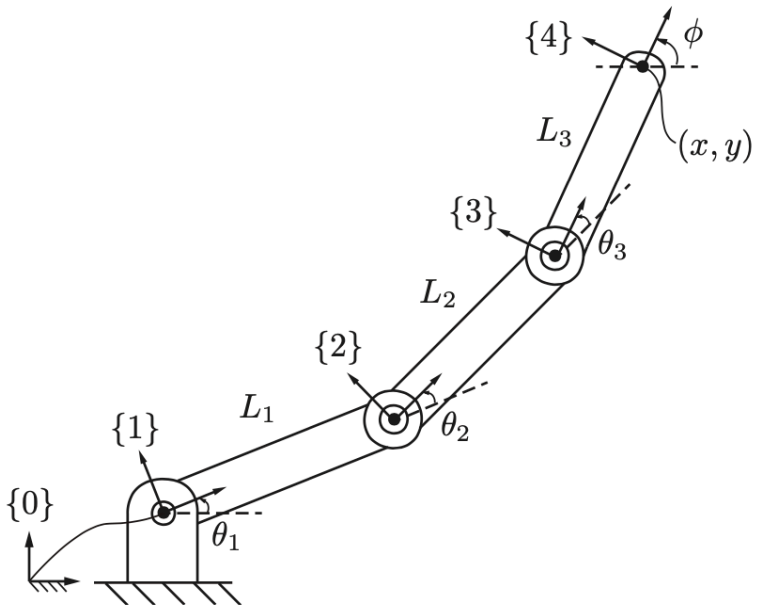
Link and Joint

Link:

- **Links** are the rigid-body connected in sequence

Joint:

- **Joints** are the connectors between links. They determine the DoF of motion between adjacent links

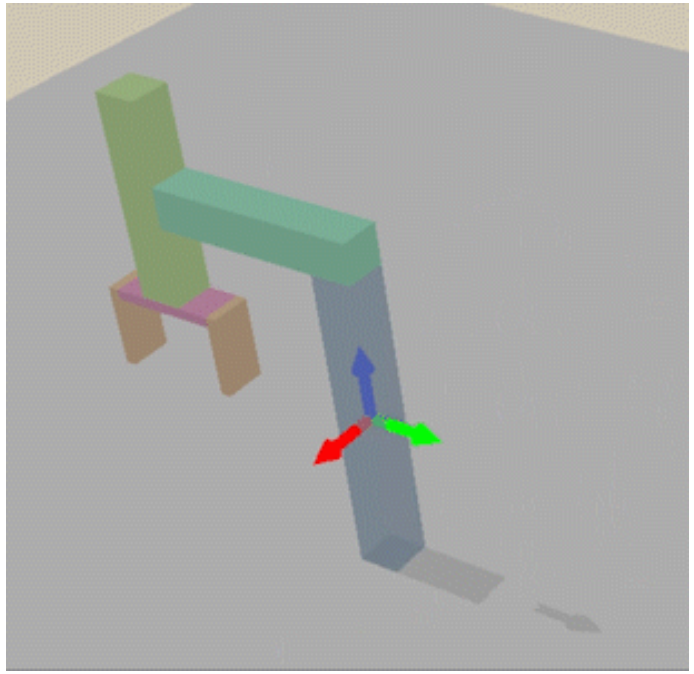


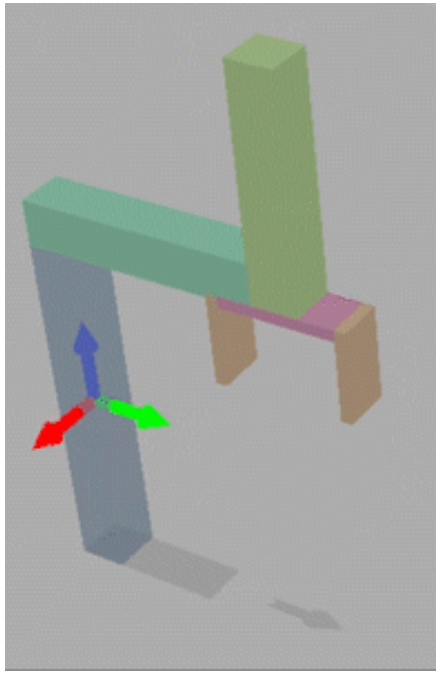
Base Link and End-Effector Link

- Base link / root link:
  - The 0-th link of the robot
  - Regarded as the “fixed” reference
  - The spatial frame $\mathscr{F}_s$ is attached to it

Examples

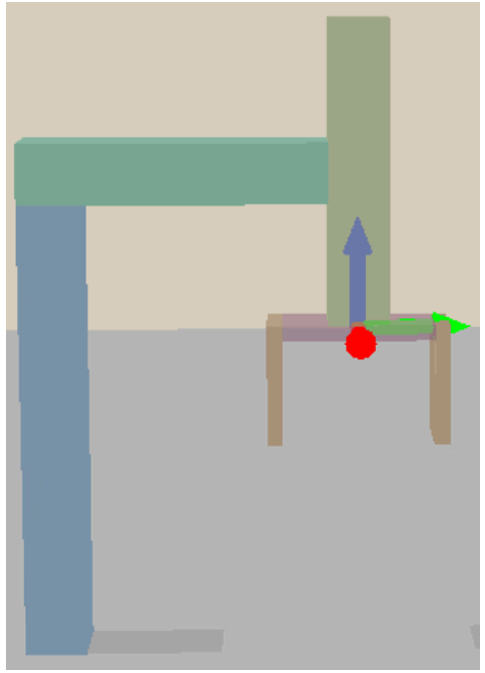
A simple 2 DoF robot arm



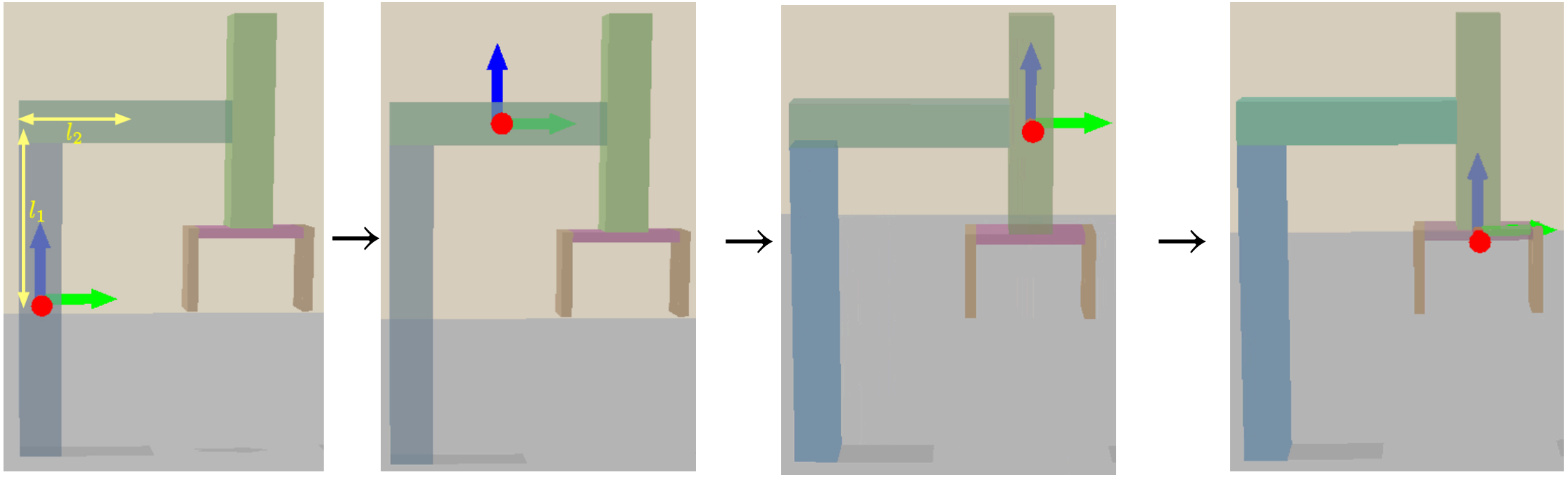


Base Link and End-Effector Link

- Base link / root link:
  - The 0-th link of the robot
  - Regarded as the “fixed” reference
  - The spatial frame $\mathscr{F}_s$ is attached to it
- End-effector link
  - The last link
  - e.g., the gripper
  - A frame $\mathscr{F}_e$ is attached to it



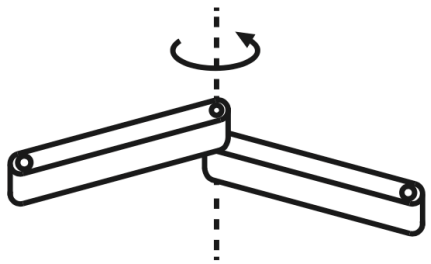
Examples



base → link1 → link2 → end_effector

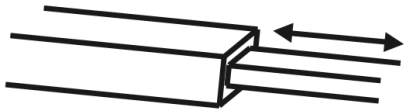
Two Common Joint Types

- Revolute/Hinge/Rotational joint



Revolute
(R)

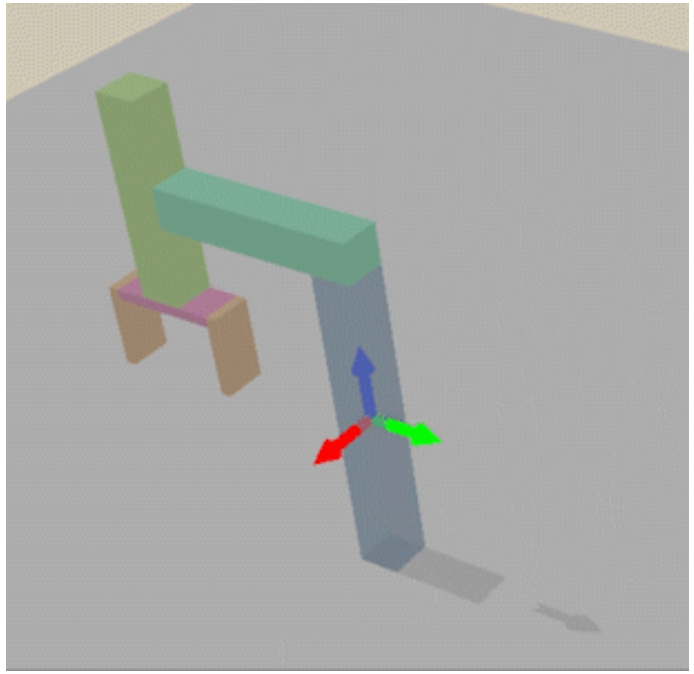
- Prismatic/Translational joint



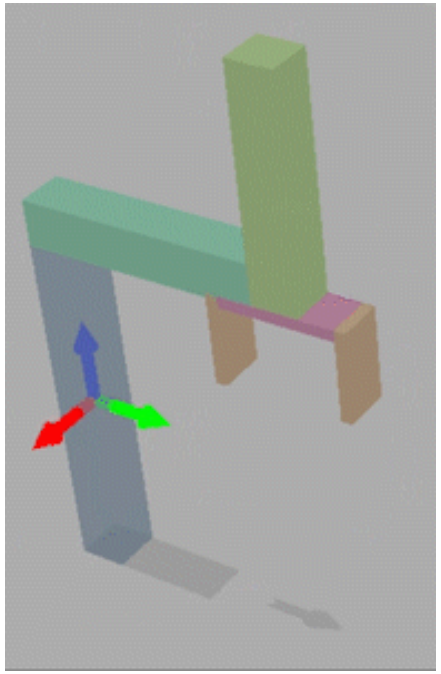
Prismatic
(P)

Examples

A simple 2 DoF robot arm



revolute (θ_1)



prismatic (θ_2)

Kinematics Configuration

- Assuming frames are assigned to each link, we can parameterize **the pose of each joint**
  - Using the relative **angle** and **translation** between adjacent frames
- Two representations of the pose of the end-effector
  - **Joint space:** The space in which each coordinate is a vector of joint poses (**angles** around **joint axis**)
  - **Cartesian space:** The space of the rigid transformations of the end-effector by $(R_{s\to e}, \mathbf{t}_{s\to e})$, where $\mathscr{F}_e$ is the end-effector frame

Forward Kinematics

- Map the joint space coordinate $\theta \in \mathbb{R}^n$ to a transformation matrix T:

$$T_{s \to e} = f(\theta)$$

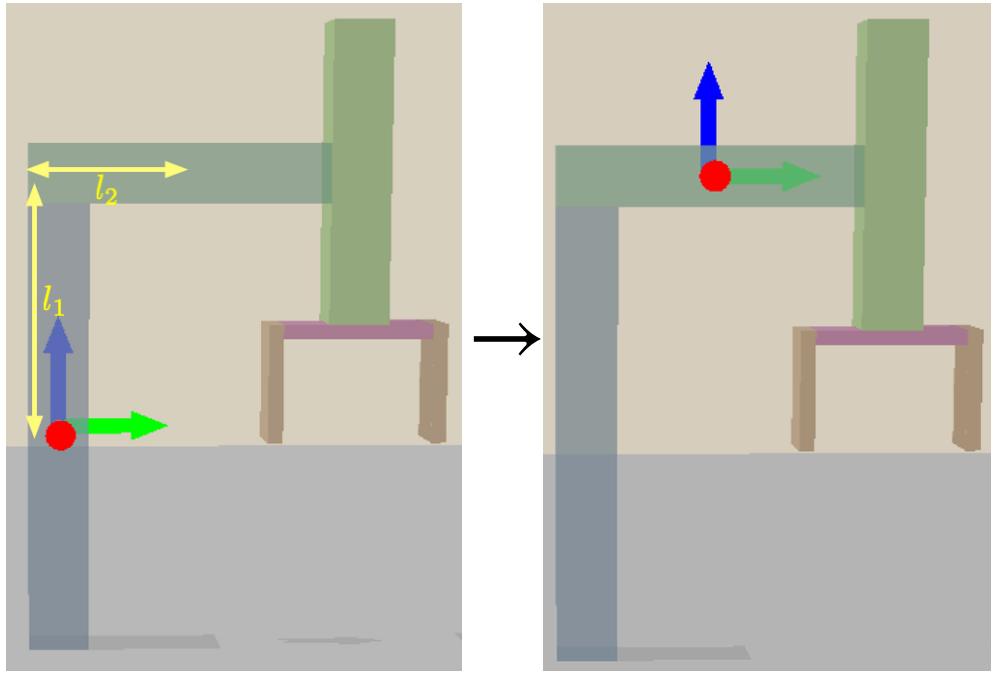
- Calculated by composing transformations along the kinematic chain

Rotation Matrix

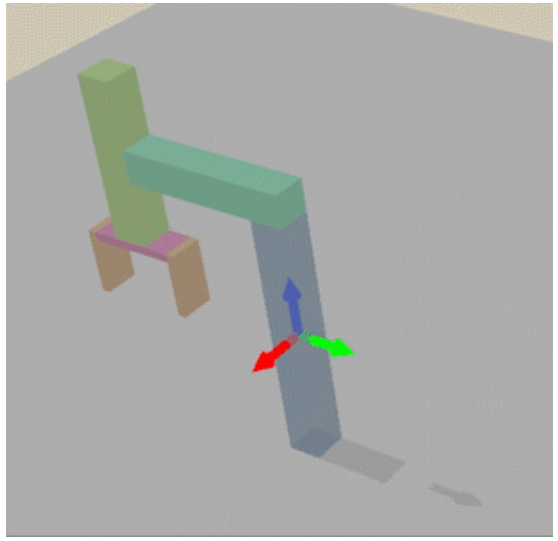
- Rotation about z axis is represented as:

$$R_z(\gamma) := \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Examples



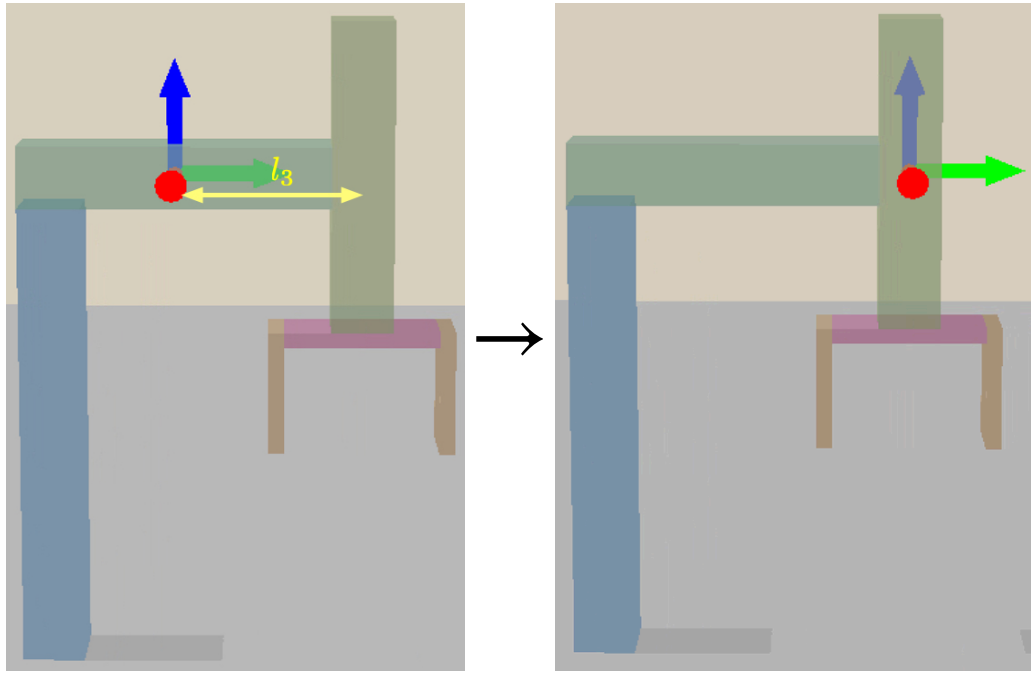
base → link1



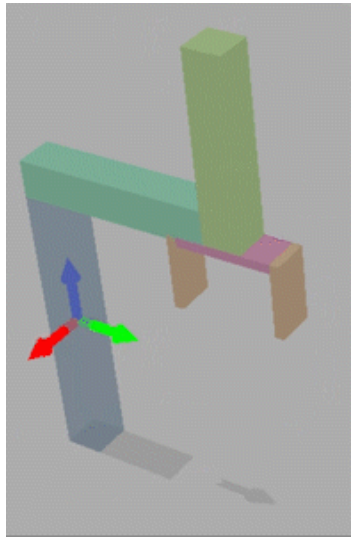
revolute (θ_1)

$$T^0_{0\to 1} = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & -l_2\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1 & 0 & l_2\cos\theta_1 \\ 0 & 0 & 1 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Examples



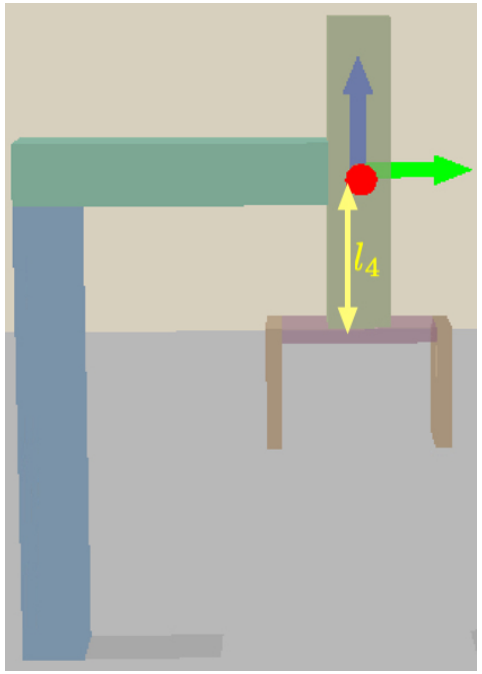
link1 → link2



prismatic (θ_2)

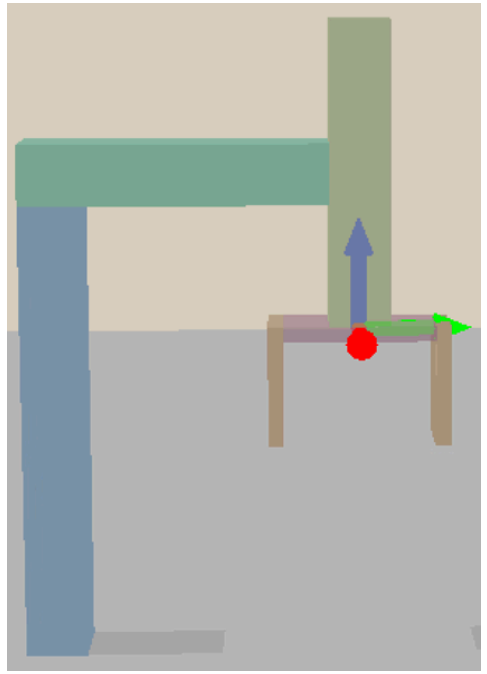
$$T^1_{1\to 2} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & l_3 \\ 0 & 0 & 1 & \theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Examples



link2

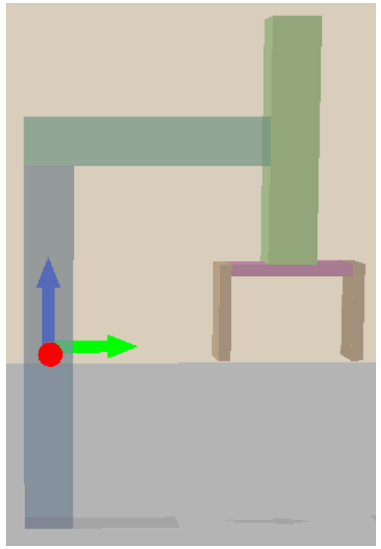
$\rightarrow$



end_effector

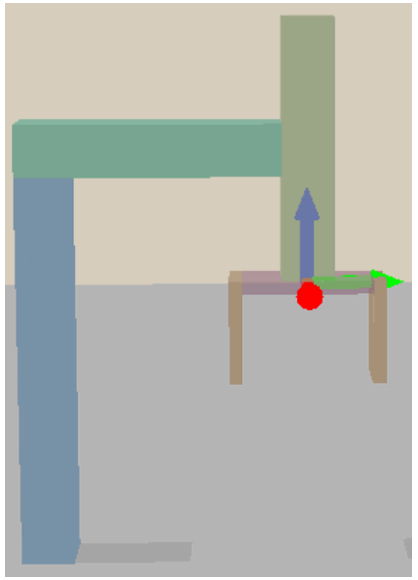
$$T^2_{2\rightarrow 3} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -l_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Kinematic Equations



base

→



end_effector

$$T^0_{0\to3} = T^0_{0\to1}T^1_{1\to2}T^2_{2\to3} = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & -\sin\theta_1(l_2+l_3) \\ \sin\theta_1 & \cos\theta_1 & 0 & \cos\theta_1(l_2+l_3) \\ 0 & 0 & 1 & l_1 - l_4 + \theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} R^s_{s\to e} & \mathbf{t}^s_{s\to e} \\ 0 & 1 \end{bmatrix}$$

Inverse Kinematics

- Given the forward kinematics $T_{s \to e}(\theta)$ and the target pose $T_{target} = \mathbb{SE}(3)$, find solutions θ that satisfy $T_{s \to e}(\theta) = T_{target}$

How to Relate the Motion in Joint Space and Cartesian Space?

- Q1: If the robot moves by $\Delta\theta$ in the joint space, how much will it move in the Cartesian space? (Forward Kinematics)
- Q2: If the robot would move the end-effector by Δx in the Cartesian space, how shall it change its joint poses? (Inverse Kinematics)
- Suppose we parameterize (R, t) by the **angles** around **axis**, we need to derive the differential map

$\mathbb{SO}(3)$ and $\mathbb{SE}(3)$

Rotation in $\mathbb{R}^3$

$\mathbb{SO}(3)$: The Space of Rotations

- $\mathbb{SO}(n) = \{R \in \mathbb{R}^{n \times n} : \det(R) = 1, RR^T = I\}$
- $\mathbb{SO}(n)$: "Special Orthogonal Group"

- "Group": roughly, closed under matrix multiplication
- "Orthogonal": $RR^T = I$
- "Special": $\det(R) = 1$

- $\mathbb{SO}(2)$: 2D rotations, 1 DoF
- $\mathbb{SO}(3)$: 3D rotations, 3 DoF

$\mathbb{SE}(3)$: The Space of Rigid Transformations

- $\mathbb{SE}(3) := \left\{ T = \begin{bmatrix} R & \mathbf{t} \\ 0 & 1 \end{bmatrix}, R \in \mathbb{SO}(3), \mathbf{t} \in \mathbb{R}^3 \right\}$
- $\mathbb{SE}(3)$: "Special Euclidean Group"

- "Group": roughly, closed under matrix multiplication
- "Euclidean": R and $\mathbf{t}$
- "Special": $\det(R) = 1$

- 6 DoF

$\mathbb{SO}(3)$ and $\mathbb{SE}(3)$

- We need some theoretical understanding of $\mathbb{SO}(3)$ and $\mathbb{SE}(3)$
  - The parameterization

Parameterization

- Record $R \in \mathbb{SO}(3)$ by real numbers
- In other words, we look for mappings between $\mathbb{R}^d$ and $\mathbb{SO}(3)$:

$$f(\theta) = R_\theta$$

Euler Angle

Euler Angle is Very Intuitive

Euler Angle to Rotation Matrix

- Rotation about principal axis is represented as:

$$R_x(\alpha) := \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix}$$

$$R_y(\beta) := \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix}$$

$$R_z(\gamma) := \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- $R = R_z(\alpha)R_y(\beta)R_x(\gamma)$ for arbitrary rotation

Inspection

- Euler Angle is not unique for some rotations.
- For example,

$$R_z(45^\circ)R_y(90^\circ)R_x(45^\circ) = R_z(90^\circ)R_y(90^\circ)R_x(90^\circ)$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

Gimbal Lock

- For example: When $\beta = \pi/2$,

$$R = R_z(\alpha)R_y(\pi/2)R_x(\gamma)$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ \sin(\alpha+\gamma) & \cos(\alpha+\gamma) & 0 \\ -\cos(\alpha+\gamma) & \sin(\alpha+\gamma) & 0 \end{bmatrix}$$

since changing α and γ has the same effects, a degree of freedom disappears!

Summary

- Euler angle can parameterize every rotation and has good interpretability
- It is not a unique representation at some points
- There are some points where not every change in the target space (rotations) can be realized by a change in the source space (Euler angles)

Angle-Axis Parameterization of Rotations

Euler's Theorem

- Any rotation is equivalent to a rotation about a fixed axis $\hat{\omega} \in \mathbb{R}^3$ ($\|\hat{\omega}\| = 1$) through a positive angle θ
- $\hat{\omega}$: unit vector of rotation axis
- θ: angle of rotation
- $R \in \mathbb{SO}(3) := \mathrm{Rot}(\hat{\omega}, \theta)$

Given $\hat{\omega}$ and θ, what is $R \in \mathbb{SO}(3)$?

Skew-Symmetric Matrix

- A is skew-symmetric $A = -A^T$
- Skew-symmetric matrix operator:

$$a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}, [a] := \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}$$

- Cross product can be a linear transformation
  - $a \times b = [a]b$

Given $\hat{\omega}$ and θ, what is $R \in \mathbb{SO}(3)$?

- We can show that, for any $x \in \mathbb{R}^3$

$$\begin{aligned}\mathrm{Rot}(\hat{\omega}, \theta)x &= x + (\sin\theta)\hat{\omega} \times x + (1 - \cos\theta)\hat{\omega} \times (\hat{\omega} \times x) \\ &= \{I + [\hat{\omega}]\sin\theta + [\hat{\omega}]^2(1 - \cos\theta)\}x \qquad (1)\end{aligned}$$

Given $\hat{\omega}$ and θ, what is $R \in \mathbb{SO}(3)$?

- We can show that, for any $x \in \mathbb{R}^3$
$$\mathrm{Rot}(\hat{\omega}, \theta)x = x + (\sin\theta)\hat{\omega} \times x + (1 - \cos\theta)\hat{\omega} \times (\hat{\omega} \times x)$$
$$= \{I + [\hat{\omega}]\sin\theta + [\hat{\omega}]^2(1 - \cos\theta)\}x \qquad (1)$$
- By Taylor's expansion of **sin**, **cos**, $[\hat{\omega}]^3 = -[\hat{\omega}]$, and above
$$\mathrm{Rot}(\hat{\omega}, \theta)x = (I + \theta[\hat{\omega}] + \frac{\theta^2}{2!}[\hat{\omega}]^2 + \frac{\theta^3}{3!}[\hat{\omega}]^3 + \cdots)x$$

Given $\hat{\omega}$ and θ, what is $R \in \mathbb{SO}(3)$?

- We can show that, for any $x \in \mathbb{R}^3$
$$\mathrm{Rot}(\hat{\omega}, \theta)x = x + (\sin\theta)\hat{\omega} \times x + (1 - \cos\theta)\hat{\omega} \times (\hat{\omega} \times x)$$
$$= \{I + [\hat{\omega}]\sin\theta + [\hat{\omega}]^2(1 - \cos\theta)\}x \qquad (1)$$
- By Taylor's expansion of **sin**, **cos**, $[\hat{\omega}]^3 = -[\hat{\omega}]$, and above
$$\mathrm{Rot}(\hat{\omega}, \theta)x = (I + \theta[\hat{\omega}] + \frac{\theta^2}{2!}[\hat{\omega}]^2 + \frac{\theta^3}{3!}[\hat{\omega}]^3 + \cdots)x$$
- Recall Taylor's expansion of exponential,
$$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots$$

Given $\hat{\omega}$ and θ, what is $R \in \mathbb{SO}(3)$?

- We can show that, for any $x \in \mathbb{R}^3$
$$\mathrm{Rot}(\hat{\omega}, \theta)x = x + (\sin\theta)\hat{\omega} \times x + (1 - \cos\theta)\hat{\omega} \times (\hat{\omega} \times x)$$
$$= \{I + [\hat{\omega}]\sin\theta + [\hat{\omega}]^2(1 - \cos\theta)\}x \qquad (1)$$
- By Taylor's expansion of **sin**, **cos**, $[\hat{\omega}]^3 = -[\hat{\omega}]$, and above
$$\mathrm{Rot}(\hat{\omega}, \theta)x = (I + \theta[\hat{\omega}] + \frac{\theta^2}{2!}[\hat{\omega}]^2 + \frac{\theta^3}{3!}[\hat{\omega}]^3 + \cdots)x$$
- Recall Taylor's expansion of exponential,
$$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots$$
- Formally, we have:
$$\mathrm{Rot}(\hat{\omega}, \theta)x = e^{[\hat{\omega}]\theta}x, \forall x \in \mathbb{R}^3$$

Given $\hat{\omega}$ and θ, what is $R \in \mathbb{SO}(3)$?

- By $\mathrm{Rot}(\hat{\omega}, \theta)x = e^{[\hat{\omega}]\theta}x, \forall x \in \mathbb{R}^3$,

$$\mathrm{Rot}(\hat{\omega}, \theta) \equiv e^{[\hat{\omega}]\theta}$$

- This is under such a **Definition of Matrix Exponential**:

$$e^{[\hat{\omega}]\theta} = I + \theta[\hat{\omega}] + \frac{\theta^2}{2!}[\hat{\omega}]^2 + \frac{\theta^3}{3!}[\hat{\omega}]^3 + \cdots$$

- By (1) in the last slide, we obtain **Rodrigues** formula:

$$e^{[\hat{\omega}]\theta} = I + [\hat{\omega}]\sin\theta + [\hat{\omega}]^2(1 - \cos\theta)$$

Given $\hat{\omega}$ and θ, what is $R \in \mathbb{SO}(3)$?

- In the angle-axis representation of $\mathrm{Rot}(\hat{\omega}, \theta) = e^{[\hat{\omega}]\theta}$
- $\vec{\theta} = \hat{\omega}\theta$ is also called the **rotation vector**, or **exponential coordinate**

Rodrigues Formula

- Definition of Matrix Exponential:

$$e^{[\hat{\omega}]\theta} = I + \theta[\hat{\omega}] + \frac{\theta^2}{2!}[\hat{\omega}]^2 + \frac{\theta^3}{3!}[\hat{\omega}]^3 + \cdots$$

- Sum of infinite series? **Rodrigues Formula**
  - Can prove that $[\hat{\omega}]^3 = -[\hat{\omega}]$
  - Then, use Taylor expansion of **sin** and **cos**
  - $e^{[\hat{\omega}]\theta} = I + [\hat{\omega}]\sin\theta + [\hat{\omega}]^2(1 - \cos\theta)$

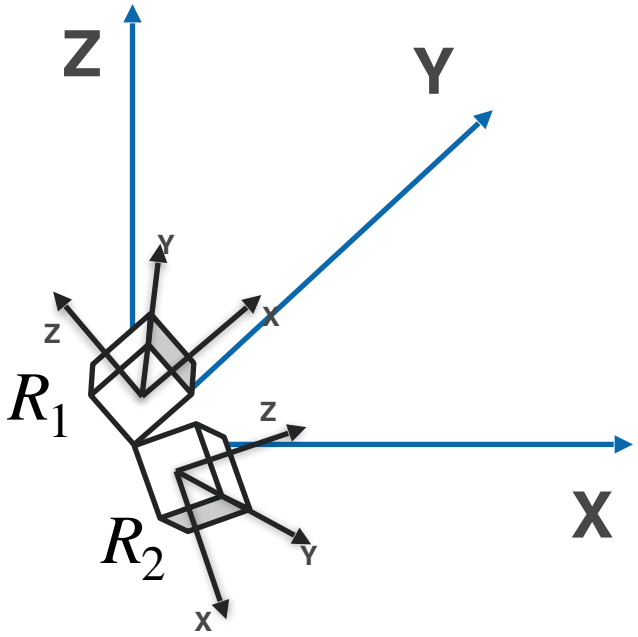
Given $\hat{\omega}$ and θ, what is $R \in \mathbb{SO}(3)$?

- First question: Is there a **unique** parametrization?
  - No:
    1. $(\hat{\omega}, \theta)$ and $(-\hat{\omega}, -\theta)$ give the same rotation
    2. when $R = I$, $\theta = 0$ and $\hat{\omega}$ can be arbitrary
    3. $(\hat{\omega}, \pi)$ and $(-\hat{\omega}, \pi)$ give the same rotation $(\mathrm{tr}(R) = -1)$
- If we restrict $\theta \in (0,\pi)$, a unique parameterization exists:
  - $\theta = \arccos \frac{1}{2}[\mathrm{tr}(R) - 1], \quad [\hat{\omega}] = \frac{1}{2\sin\theta}(R - R^T)$

Distance between Rotations

- How to measure the distance between rotations (R_1, R_2)?
- A natural view is to measure the (minimal) effort to rotate the body at R_1 pose to R_2 pose:

$$\because (R_2 R_1^T) R_1 = R_2 \qquad \therefore \mathrm{dist}(R_1, R_2) = \theta(R_2 R_1^T) = \arccos \frac{1}{2}[\mathrm{tr}(R_2 R_1^T) - 1]$$



Quaternion

Note: In this section, $\vec{x} \in \mathbb{R}^3$ and $q \in \mathbb{R}^4$

Quaternion is a "Number"

- Recall the complex number $a + b\mathbf{i}$
- Quaternion is a more generalized complex number:

$$q = w + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

  - w is the real part and $\vec{v} = (x, y, z)$ is the imaginary part
  - Imaginary: $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$
  - anti-commutative : $\mathbf{ij} = \mathbf{k} = -\mathbf{ji},\ \mathbf{jk} = \mathbf{i} = -\mathbf{kj},\ \mathbf{ki} = \mathbf{j} = -\mathbf{ik}$

Properties of General Quaternions

- Vector form: $q = (w, \vec{v})$
- Product:
  - For $q_1 = (w_1, \vec{v}_1)$ and $q_2 = (w_2, \vec{v}_2)$,
    $q_1 q_2 = (w_1 w_2 - \vec{v}_1^T \vec{v}_2, w_1 \vec{v}_2 + w_2 \vec{v}_1 + \vec{v}_1 \times \vec{v}_2)$
  - Not commutable (note that $\vec{v}_1 \times \vec{v}_2 \neq \vec{v}_2 \times \vec{v}_1$)
- Conjugate: $q^* = (w, -\vec{v})$
- Norm: $\|q\|^2 = w^2 + \vec{v}^T \vec{v} = qq^* = q^* q$
- Inverse: $q^{-1} := \dfrac{q^*}{\|q\|^2}$

Unit Quaternion as Rotation

- A **unit** quaternion $\|\boldsymbol{q}\| = 1$ can represent a rotation
  - Four numbers plus one constraint $\rightarrow$ 3 DoF
- Geometrically, the shell of a 4D sphere

Build Rotation Quaternion

- Exponential coordinate → Quaternion:

$$q = [\cos(\theta/2), \sin(\theta/2)\hat{\omega}]$$

- Quaternion is very close to angle-axis representation!
- Exponential coordinate ← Quaternion: $\theta = 2\arccos(w)$,

$$\hat{\omega} = \begin{cases} \dfrac{1}{\sin(\theta/2)}\vec{v} & \theta \neq 0 \\ 0 & \theta = 0 \end{cases}$$

Unit Quaternion as Rotation

- Rotate a vector $\vec{x}$ by a quaternion q:
  1. Augment $\vec{x}$ to $x = (0,\vec{x})$
  2. $x' = qxq^{-1}$
- Compose rotations by quaternion:
  - $(q_2(q_1xq_1^*)q_2^*)$: first rotate by q_1 and then by q_2
  - Since $(q_2(q_1xq_1^*)q_2^*) = (q_2q_1)x(q_1^*q_2^*)$, composing rotations is as simple as multiplying quaternions!

Next week: Lecture 3, Robotics II